SECTION 2: Density & Unit Cell Calculations (Q41–80)

41. A metal crystallizes in FCC structure with edge length 4 Å. Atomic mass = 64 g/mol. Calculate density.
    A) 8 g/cm^3
    B) 7.2 g/cm^3
    C) 8.96 g/cm^3
    D) 9 g/cm^3
42. SC lattice: edge length = 3 Å, atomic mass = 50 g/mol. Density = ?
    A) 4.65 g/cm^3
    B) 5.55 g/cm^3
    C) 6 g/cm^3
    D) 3.5 g/cm^3
43. A BCC metal has density 8 g/cm^3, atomic mass 55.8 g/mol. Edge length = ?
    A) 2.87 Å
    B) 3 Å
    C) 3.2 Å
    D) 4 Å
44. Atomic radius of FCC metal with edge length 4.05 Å = ?
    A) 1.43 Å
    B) 1 Å
    C) 2 Å
    D) 1.5 Å
45. NaCl unit cell: edge length = 5.64 Å. Find density.
    A) 2.17 g/cm^3
    B) 2.16 g/cm^3
    C) 2 g/cm^3
    D) 2.5 g/cm^3
46. CsCl unit cell: edge length = 4.12 Å, atomic masses Cs = 133 g/mol, Cl = 35.5 g/mol. Density = ?
    A) 3.99 g/cm^3
    B) 4.01 g/cm^3
    C) 4.2 g/cm^3
    D) 4.5 g/cm^3
47. Fraction of volume occupied by atoms in BCC lattice = ?
    A) 0.68
    B) 0.52
    C) 0.74
    D) 0.60
48. Fraction of volume occupied by atoms in SC lattice = ?
    A) 0.52
    B) 0.68
    C) 0.74
    D) 0.60
49. Number of tetrahedral voids in FCC unit cell = ?
    A) 8
    B) 4

C) 12
D) 6

50. Number of octahedral voids in FCC unit cell = ?
A) 4
B) 6
C) 12
D) 8

51. Ionic radius ratio for stable octahedral coordination = ?
A) 0.414 – 0.732
B) 0.225 – 0.414
C) 0.732 – 1
D) 0.155 – 0.225

52. Ionic radius ratio for stable tetrahedral coordination = ?
A) 0.225 – 0.414
B) 0.414 – 0.732
C) 0.732 – 1
D) 0.155 – 0.225

53. Density of HCP metal with 2 atoms/unit cell, molar mass = 27 g/mol, atomic radius = 1.43 Å = ?
A) 2.7 g/cm^3
B) 2.65 g/cm^3
C) 2.6 g/cm^3
D) 2.5 g/cm^3

54. Distance between nearest neighbors in FCC lattice = ?
A) a/√2
B) a
C) a√2
D) a/2

55. Distance between nearest neighbors in BCC lattice = ?
A) a√3/2
B) a/√2
C) a
D) a/2

56. Mass of atoms per unit cell in FCC = ?
A) 4 × atomic mass / N_A
B) 2 × atomic mass / N_A
C) 1 × atomic mass / N_A
D) 8 × atomic mass / N_A

57. Mass of atoms per unit cell in BCC = ?
A) 2 × atomic mass / N_A
B) 4 × atomic mass / N_A
C) 1 × atomic mass / N_A
D) 8 × atomic mass / N_A

58. Edge length of FCC lattice = 4 Å. Calculate volume of unit cell.
A) 64×10^{-24} cm^3
B) 16×10^{-24} cm^3
C) 32×10^{-24} cm^3
D) 8×10^{-24} cm^3

59. Volume of BCC unit cell with edge 3 Å = ?
    A) 27×10^{-24} cm^3
    B) 9×10^{-24} cm^3
    C) 30×10^{-24} cm^3
    D) 32×10^{-24} cm^3
60. Atomic radius of SC lattice with edge length 3 Å = ?
    A) 1.5 Å
    B) 2 Å
    C) 1 Å
    D) 1.25 Å
61. For a metal with FCC structure, the atomic packing factor = ?
    A) 0.74
    B) 0.68
    C) 0.52
    D) 0.60
62. Edge length of CsCl unit cell = 4.12 Å. Distance between ions = ?
    A) 2.06 Å
    B) 2.82 Å
    C) 2.5 Å
    D) 2 Å
63. Density of KCl (NaCl type) = ?
    A) 1.98 g/cm^3
    B) 2.0 g/cm^3
    C) 2.2 g/cm^3
    D) 2.1 g/cm^3
64. Fraction of unoccupied volume in FCC = ?
    A) 0.26
    B) 0.32
    C) 0.48
    D) 0.52
65. Fraction of unoccupied volume in BCC = ?
    A) 0.32
    B) 0.26
    C) 0.48
    D) 0.52
66. Number of formula units per unit cell for NaCl = ?
    A) 4
    B) 2
    C) 6
    D) 8
67. Number of formula units per unit cell for CsCl = ?
    A) 1
    B) 2
    C) 4
    D) 8
68. If density of a cubic metal doubles, edge length changes by factor = ?
    A) $1/\sqrt{2}$
    B) 1/2

C) √2
D) 2

69. Volume of unit cell = a^3, a = 4 Å. Number of atoms = 4 (FCC). Atomic volume = ?
A) 29.6 $Å^3$
B) 32 $Å^3$
C) 30 $Å^3$
D) 28 $Å^3$

70. Edge length of unit cell = 3.6 Å. Calculate radius of atom (BCC).
A) 1.56 Å
B) 1.43 Å
C) 1.5 Å
D) 1.2 Å

71. Ionic radius of Cl– = 1.67 Å. Cs+ in CsCl = 1.81 Å. Edge length = ?
A) 3.48 Å
B) 4.12 Å
C) 4 Å
D) 3.8 Å

72. Density of Ag (FCC) = 10.5 g/cm^3. Atomic mass = 107.8 g/mol. Edge length = ?
A) 4.08 Å
B) 4 Å
C) 3.98 Å
D) 4.2 Å

73. Number of tetrahedral voids in unit cell = 8. Edge length = 4 Å. Find volume of void.
A) 0.225 r^3
B) 0.155 r^3
C) 0.414 r^3
D) 0.732 r^3

74. Number of octahedral voids in unit cell = 4. Edge length = 4 Å. Volume of void = ?
A) 0.414 r^3
B) 0.225 r^3
C) 0.732 r^3
D) 0.155 r^3

75. Edge length of NaCl unit cell = 5.64 Å. Atomic mass Na = 23 g/mol, Cl = 35.5 g/mol. Density = ?
A) 2.16 g/cm^3
B) 2.17 g/cm^3
C) 2.0 g/cm^3
D) 2.2 g/cm^3

76. A metal crystallizes in HCP structure with 2 atoms/unit cell. Atomic mass = 27 g/mol, atomic radius = 1.43 Å. Density = ?
A) 2.65 g/cm^3
B) 2.7 g/cm^3
C) 2.6 g/cm^3
D) 2.55 g/cm^3

77. FCC metal: lattice parameter = 4 Å. Atomic mass = 64 g/mol. Calculate density.
A) 8 g/cm^3
B) 8.96 g/cm^3

C) 7.8 g/cm^3
D) 7 g/cm^3

78. Radius of atom in FCC with a = 4 Å = ?
    A) 1.414 Å
    B) 1.43 Å
    C) 1 Å
    D) 1.5 Å
79. Fraction of volume occupied by atoms in BCC = ?
    A) 0.68
    B) 0.74
    C) 0.52
    D) 0.60
80. Fraction of void in SC lattice = ?
    A) 0.48
    B) 0.52
    C) 0.32
    D) 0.26